Supplementary Materials for "The Aptitude-Achievement Paradox: A Theory of Rank Reversals under Selective Observation"

This supplementary materials include proofs of technical results and additional numerical examples supporting conclusions from Section 3 and Section 4 the main manuscript.

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1 Proofs

1.1 Proof of Theorem 2.2

Proof. Fix a population i and suppress the index i. First, since F_X is continuous, we can express well-defined order statistics as $X_{(1)} < \cdots < X_{(N)}$. By rank-based selection, the observed system consists of indices j such that $r_j \in \mathcal{R}$, yielding observed pairs $(Y_{(j)}, X_{(r_j)})$, for $j = 1, \ldots, n$. By assumption, for each observed rank j there exists a strictly increasing measurable function ψ_j such that

$$Y_{(j)} = \psi_j(X_{(r_j)}) \quad \text{a.s.}$$

We now identify the distribution function $F_{Y_{(j)}}$ for fixed j. Since ψ_j is strictly increasing, it admits a measurable inverse on its range. For any $y \in \mathbb{R}$,

$$\mathrm{P}(Y_{(j)} \leq y) = \mathrm{P}(\psi_j(X_{(r_j)}) \leq y) = \mathrm{P}(X_{(r_j)} \leq \psi_j^{-1}(y)).$$

Hence, $F_{Y_{(j)}}(y) = F_{X_{(r_j)}}(\psi_j^{-1}(y))$, which implies that

$$F_{Y_{(j)}}(\psi_j(x)) = F_{X_{(r_j)}}(x) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

We now show that $Y_{(j)}$ is recovered from the Full House Modeling construction. Since F_X is continuous, the distribution of the order statistic $X_{(r_j)}$ is also continuous and strictly increasing on the interior of its support, hence admits a well-defined generalized inverse.

Likewise, $F_{Y_{(j)}}$ is continuous and strictly increasing. Thus both admit well-defined inverses. Let $u = F_{X_{(r_j)}}(x)$. Then $F_{Y_{(j)}}(\psi_j(x)) = u$, so that

$$\psi_j(x) = F^{-1}_{Y_{(j)}}(u) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big)$$

for $F_{X_{(r_j)}}$-almost every x. Therefore,

$$Y_{(j)} = F^{-1}_{Y_{(j)}} \circ F_{X_{(r_j)}}(X_{(r_j)}) \quad \text{a.s.}$$

We now verify the uniqueness of the above. Let $U = F_{X_{(r_j)}}(X_{(r_j)}) \sim \text{Unif}(0,1)$ by the probability integral transform, and define

$$g(u) := \psi_j\big(F^{-1}_{X_{(r_j)}}(u)\big).$$

Then g is strictly increasing and $g(U) \sim F_{Y_{(j)}}$. By the fact that if $U \sim \text{Unif}(0,1)$ and g is increasing, then the distribution of $g(U)$ uniquely determines g almost everywhere, implying $g(u) = F^{-1}_{Y_{(j)}}(u)$ a.e. It follows that

$$\psi_j(x) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

Thus, the strictly increasing coupling is unique up to null sets. The representation holds for each observed rank j, and uniqueness implies that the joint distribution of (X, Y) coincides with that induced by the Full House Model. □

1.2 Proof of Lemma 3.2

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right)$$

$$= \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right)$$
$$= \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = (f_{1,j}^{-1} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

1.3 Proof of Theorem 3.4

Proof. By Lemma 3.2,

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Under the assumption $F_{Y_1} = F_{Y_2} = F_Y$, the distributions of the order statistics admit Beta representations, which allow us to express the mapping $h_{j,k}$ in terms of regularized incomplete beta functions.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} \\
&\quad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x)} d\, I_{r_{1,j},N_1-r_{1,j}+1}(F_X(y)) \right\} \\
&\quad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \\
&\quad \times f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \\
&\quad d\, I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x))
\end{aligned}$$

$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

This representation holds without additional assumptions. The positive-part operator appears because the integrand may change sign. Under condition (8), the integrand is nonnegative, hence

$$\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} dt.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, dI_{r_{2,k},N_2-r_{2,k}+1}(x) \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(k, n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

□

1.4 Proof of Proposition 3.5

Proof. Specification of the selection rule $S_{i,j} = 1 \iff X_{i,j} \geq X_{i,(N_i - n_i + 1)}$ implies that $r_{i,j} = N_i - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $t \in (0, 1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased. This follows from the monotonicity of the inverse incomplete beta function in its parameters [Askitis, 2021], together with the fact that stochastic dominance is preserved under monotone transformations. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned}
&I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\
&\quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\
&\quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq 0.
\end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

1.5 Proof of Lemma 3.6

Proof. The inverse-function derivative gives

$$q_s'(u) = \frac{B(s-j, j+1)}{q_s(u)^{s-j-1}(1-q_s(u))^j} = \frac{q_s(u)}{s h_s(u)}.$$

Consequently,

$$h_s'(u) = 1 - \frac{j}{s(1-q_s(u))}, \qquad h_s''(u) = -\frac{j q_s(u)}{s^2(1-q_s(u))^2 h_s(u)}.$$

Since $s - j > 0$ and $j > 0$, h_s extends continuously to $[0, 1]$ with $h_s(0) = h_s(1) = 0$.

Put $m = h_n - h_N$. If m were nonnegative at any point of $(0,1)$, then it would attain a nonnegative maximum at some $u_0 \in (0,1)$. At this point,

$$m(u_0) \geq 0, \qquad m'(u_0) = 0, \qquad m''(u_0) \leq 0.$$

The equality $m'(u_0) = 0$ implies

$$n(1 - q_n(u_0)) = N(1 - q_N(u_0)) =: c > 0,$$

and hence

$$q_N(u_0) - q_n(u_0) = c\left(\frac{1}{n} - \frac{1}{N}\right) > 0.$$

Because $h_n(u_0) \geq h_N(u_0) > 0$, it follows that

$$\begin{aligned} m''(u_0) &= \frac{j}{c^2}\left(\frac{q_N(u_0)}{h_N(u_0)} - \frac{q_n(u_0)}{h_n(u_0)}\right) \\ &\geq \frac{j}{c^2}\,\frac{q_N(u_0) - q_n(u_0)}{h_n(u_0)} > 0, \end{aligned}$$

a contradiction. Thus $m(u) < 0$ for all $u \in (0,1)$. □

1.6 Proof of Proposition 3.7

Proof. Specification of the selection rule $S_{i,j} = 1 \iff X_{i,j} \geq X_{i,(N_i - n_i + 1)}$ implies that $r_{i,j} = N_i - j$. Using the reduction of technical condition (8) for this model, it suffices to show that

$$f(x) := I_{n_1 - j, j+1}\big(I^{-1}_{n_2 - j, j+1}(x)\big) - I_{N_1 - j, j+1}\big(I^{-1}_{N_2 - j, j+1}(x)\big) \geq 0 \qquad (0 < x < 1).$$

If $j = 0$, then $I_{a,1}(t) = t^a$ and

$$f(x) = x^{n_1/n_2} - x^{N_1/N_2} \geq 0,$$

because the ratio assumption gives $n_1/n_2 \le N_1/N_2$. Henceforth assume $j > 0$. In this case, the inequality is strict.

Let $d = n_2 - n_1$ and $D = N_2 - N_1$. The ratio assumption is equivalent to

$$\frac{d}{n_2} \ge \frac{D}{N_2}.$$

Because $D > d \ge 0$, this implies $d > 0$ and $N_2 > n_2$. It also implies $N_1 \ge n_1 N_2/n_2 > n_1 > j$, so all beta parameters are positive.

For $s \in \{n_2, N_2\}$, use the notation

$$q_s(x) = I^{-1}_{s-j,j+1}(x), \qquad h_s(x) = \frac{q_s(x)^{s-j}(1-q_s(x))^j}{sB(s-j,j+1)}.$$

Define the positive constants

$$A = \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)}, \qquad C = \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)}.$$

Differentiating the two compositions yields

$$f'(x) = Aq_{n_2}(x)^{-d} - Cq_{N_2}(x)^{-D} = q_{n_2}(x)^{-d}\big(A - Cr(x)\big),$$

where

$$r(x) = \frac{q_{n_2}(x)^d}{q_{N_2}(x)^D}.$$

By Lemma 3.6, $h_{n_2}(x) < h_{N_2}(x)$. Therefore

$$\begin{aligned}\frac{r'(x)}{r(x)} &= \frac{d}{n_2 h_{n_2}(x)} - \frac{D}{N_2 h_{N_2}(x)} \\ &\ge \frac{d}{n_2}\left(\frac{1}{h_{n_2}(x)} - \frac{1}{h_{N_2}(x)}\right) > 0.\end{aligned}$$

Thus r is strictly increasing, and $A - Cr(x)$ is strictly decreasing.

The function f extends continuously to $[0,1]$ with $f(0) = f(1) = 0$. By Rolle's theorem, there exists $x_* \in (0,1)$ such that $f'(x_*) = 0$, equivalently $r(x_*) = A/C$. Strict monotonicity of r makes this point unique and gives

$$f'(x) > 0 \quad (0 < x < x_*), \qquad f'(x) < 0 \quad (x_* < x < 1).$$

Together with $f(0) = f(1) = 0$, these inequalities show that $f(x) > 0$ for every $x \in (0,1)$, which establishes the required beta-function inequality. □

1.7 Proof of Proposition 4.2

Proof. First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{+\infty} \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt.
\end{aligned}$$

Based on the Definition 4.1, we have

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt
\end{aligned}$$

$$= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},$$

and

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\leq \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}$$

□

1.8 Proof of Theorem 4.4

Proof. Equation 7 from Lemma 3.2 yields

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2N_1}{N_2n_1}}\right)\sigma_1 + \mu_1.$$

Let A and R be as defined, respectively, in (16) and (15). We have,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad = \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2N_1}{N_2n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad = \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&\quad = \int_A \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) \\
&\quad = \int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) - \int_{A^c} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) \\
&\quad = \int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) + R.
\end{aligned}$$

We continue with the left-hand side of the last equality,

$$\begin{aligned}
\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) &= \int_0^1 \left[F_{Y_1}\big(F_{Y_2}^{-1}(t^{1/n_2})\big)^{n_1} - t^{N_1/N_2}\right] dt \\
&= \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \; - \; \frac{N_2}{N_1+N_2} \\
&= \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2 \Phi(x)^{n_2-1}\phi(x)\right) dx \; - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \int_{-\infty}^{+\infty} \mathrm{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \;\middle|\; X = x \right) \phi(x)\, dx \\
&\quad - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \; \mathrm{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \right) \; - \; \frac{N_2}{N_1+N_2} \\
&= n_2 \, F_{\mathbf{W}}(\mathbf{w}) \; - \; \frac{N_2}{N_1+N_2},
\end{aligned}$$

where the first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last n_2-1 elements are 0. The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (18) and (19).

The bound on the remainder (17) follows from the argument in Section 4.2. □

1.9 Proof of Theorem 4.5

Proof. From (20) we have

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = n_2 F_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1+N_2} + R,$$

and we estimate the above with

$$\widehat{\mathrm{P}}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = n_2 \widehat{F}_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1 + N_2} + \widehat{R},$$

where $\widehat{F}_{\mathbf{W}}(\mathbf{w})$ and $\widehat{R}$ are obtained by replacing μ_i, σ_i with estimators $\hat{\mu}_i, \hat{\sigma}_i$, $i = 1, 2$.

By Theorem 4.4, $|R| \leq \left(\frac{1}{2}\right)^{n_2}$ under the conditions $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $\sigma_1 \leq \sigma_2$, and $\mu_1 \leq \mu_2$. These conditions are satisfied in our setting. Let $E = \{\hat{\sigma}_1 > \hat{\sigma}_2\} \cup \{\hat{\mu}_1 > \hat{\mu}_2\}$. Then $|\widehat{R}| \leq \left(\frac{1}{2}\right)^{n_2}$ on E^c. Our convergence of probability conditions guarantee that $\mathrm{P}(E^c) \to 1$ as $n_1, n_2 \to \infty$.

Now we show convergence of $\widehat{F}_{\mathbf{W}}(\mathbf{w})$. From the last equality in the proof of Theorem 4.4 we can write

$$\begin{aligned} F_{\mathbf{W}}(\mathbf{w}) &= \mathrm{P}\left(\begin{array}{l} Z_1 + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{n_1} + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \\ \\ Z_{n_1+1} + Z_0 \leq 0, \ldots, Z_{n_1+n_2-1} + Z_0 \leq 0 \end{array} \right) \\ &= \mathrm{P}\left(\begin{array}{l} Z_1 \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \ldots, Z_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \\ \\ Z_{n_1+1} \leq -Z_0, \ldots, Z_{n_1+n_2-1} \leq -Z_0 \end{array} \right) \end{aligned}$$

where $Z_i \overset{iid}{\sim} N(0,1)$, $i = 0, \ldots, n_1 + n_2 + 1$. Define $M_{n_1} = \max\{Z_1, Z_2, \ldots, Z_{n_1}\}$, $M_{n_2-1} = \max\{Z_{n_1+1}, Z_{n_1+2}, \ldots, Z_{n_1+n_2-1}\}$, then we can write

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}\left(M_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \quad M_{n_2-1} \leq -Z_0 \right).$$

From Embrechts et al. [2013], we have $M_{n_1} = \sqrt{2 \ln n_1} + O_P(1)$. Thus, after rearranging, the inequality $M_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1}$ is approximately equivalent (up to $O_P(1)$ fluctuations) to $\sqrt{2 \ln n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0$. Rearranging yields

$$Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2} \sqrt{2 \ln n_1}.$$

Similarly, from Embrechts et al. [2013], $M_{n_2-1} = \sqrt{2\ln(n_2-1)} + O_P(1)$. Thus, after rearranging, the inequality $M_{n_2-1} \le -Z_0$ is approximately equivalent to $Z_0 \le -\sqrt{2\ln(n_2-1)}$. Then, as $n_1, n_2 \to \infty$, we have

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}\left(Z_0 \le \frac{\mu_2-\mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2}\sqrt{2\ln n_1},\ \ Z_0 \le -\sqrt{2\ln(n_2-1)}\right).$$

Define

$$C := \frac{\mu_2-\mu_1}{\sigma_2} - \frac{\sigma_1}{\sigma_2}\sqrt{2\log n_1}, \qquad B := -\sqrt{2\log(n_2-1)},$$

and similarly

$$\widehat{C} = \frac{\hat\mu_2-\hat\mu_1}{\hat\sigma_2} - \frac{\hat\sigma_1}{\hat\sigma_2}\sqrt{2\log n_1}.$$

Two asymptotic regimes arise depending on whether $C < B$ or $C > B$. Under the consistency conditions

$$\frac{\hat\sigma_1}{\hat\sigma_2} - \frac{\sigma_1}{\sigma_2} = o_P\left(\frac{1}{\log n_1}\right), \qquad \frac{\hat\mu_2-\hat\mu_1}{\hat\sigma_2} - \frac{\mu_2-\mu_1}{\sigma_2} = o_P\left(\frac{1}{\sqrt{\log n_1}}\right),$$

and the regime-separation condition

$$\sqrt{\log n_1}\,|C-B| \to \infty,$$

the plug-in estimator selects the same regime with probability tending to one.

Indeed,

$$\widehat{C} - C = \left\{\frac{\hat\mu_2-\hat\mu_1}{\hat\sigma_2} - \frac{\mu_2-\mu_1}{\sigma_2}\right\} - \sqrt{2\log n_1}\left\{\frac{\hat\sigma_1}{\hat\sigma_2} - \frac{\sigma_1}{\sigma_2}\right\} = o_P\left(\frac{1}{\sqrt{\log n_1}}\right).$$

Therefore, if $C < B$, then

$$P(\widehat{C} > B) = P(\widehat{C} - C > B - C) \to 0,$$

because $B - C \gg 1/\sqrt{\log n_1}$. Similarly, if $C > B$, then

$$P(\widehat{C} < B) = P(C - \widehat{C} > C - B) \to 0.$$

Hence the plug-in estimator selects the same regime as the population quantity with probability tending to one.

First, consider the asymptotic regime $C \leq B$ where

$$F_{\mathbf{W}}(\mathbf{w}) = \mathrm{P}(Z_0 \leq C)\big(1 + o(1)\big) = \Phi(C)\big(1 + o(1)\big).$$

Since $C \to -\infty$, Mills' ratio $\Phi(-x) \sim \phi(x)/x$ gives

$$\Phi(C) \;\sim\; \frac{1}{\sqrt{2\pi}\,|C|}\,\exp\!\left(-\tfrac{1}{2}\left(\tfrac{\mu_2-\mu_1}{\sigma_2} - \tfrac{\sigma_1}{\sigma_2}\sqrt{2\ln n_1}\right)^2\right).$$

Let $\widehat{F}_{\mathbf{W}}(\mathbf{w})$ be obtained by replacing μ_i, σ_i with consistent estimators $\hat{\mu}_i, \hat{\sigma}_i$, and define

$$\widehat{C} := \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_2} - \frac{\hat{\sigma}_1}{\hat{\sigma}_2}\sqrt{2\ln n_1}.$$

Replacing C with $\widehat{C}$ yields

$$\widehat{F}_{\mathbf{W}}(\mathbf{w}) \;\sim\; \frac{1}{\sqrt{2\pi}\,|\widehat{C}|}\,\exp\!\left(-\tfrac{1}{2}\left(\tfrac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2} - \tfrac{\hat{\sigma}_1}{\hat{\sigma}_2}\sqrt{2\ln n_1}\right)^2\right),$$

and therefore

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \;\sim\; \frac{|C|}{|\widehat{C}|}\,\exp\!\Big(-\tfrac{1}{2}\big(\widehat{C}^{\,2} - C^2\big)\Big).$$

Writing out the squares,

$$\widehat{C}^{\,2} - C^2 = \underbrace{\left[\left(\tfrac{\hat{\mu}_2-\hat{\mu}_1}{\hat{\sigma}_2}\right)^2 - \left(\tfrac{\mu_2-\mu_1}{\sigma_2}\right)^2\right]}_{o_P(1)}$$

$$+ (2\ln n_1)\left[\frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} - \frac{\sigma_1^2}{\sigma_2^2}\right] - 2\sqrt{2\ln n_1}\left[\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_2} \cdot \frac{\hat{\sigma}_1}{\hat{\sigma}_2} - \frac{\mu_2 - \mu_1}{\sigma_2} \cdot \frac{\sigma_1}{\sigma_2}\right].$$

The first bracket is $o_P(1)$ under any consistent estimator. The remaining two terms grow with n_1 at rates $\ln n_1$ and $\sqrt{\ln n_1}$, so they govern the asymptotic behavior of $\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})}$.

Provided $\hat{\sigma}_1/\hat{\sigma}_2 \xrightarrow{P} \sigma_1/\sigma_2$, the prefactor $|C|/|\widehat{C}|$ tends to 1 in probability. Combined with the exponent expansion,

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \xrightarrow{P} 1$$

whenever the assumed conditions,

$$(2\ln n_1)\left|\frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} - \frac{\sigma_1^2}{\sigma_2^2}\right| \xrightarrow{P} 0,$$
$$\sqrt{2\ln n_1}\left|\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_2} \cdot \frac{\hat{\sigma}_1}{\hat{\sigma}_2} - \frac{\mu_2 - \mu_1}{\sigma_2} \cdot \frac{\sigma_1}{\sigma_2}\right| \xrightarrow{P} 0,$$

are satisfied.

Now consider the regime $B \le C$. In this case we have $F_{\mathbf{W}}(\mathbf{w}) = \Phi(B)(1 + o(1))$. Since $B = -\sqrt{2\log(n_2 - 1)}$, Mills' ratio yields

$$F_{\mathbf{W}}(\mathbf{w}) \sim \frac{1}{\sqrt{4\pi\log(n_2 - 1)}} \cdot \frac{1}{n_2 - 1}.$$

Under consistency of the plug-in estimators, $\widehat{C} - C \xrightarrow{P} 0$, so whenever the inequality $B < C$ holds asymptotically with nonvanishing separation, the plug-in estimator selects the same regime with probability tending to one. Consequently, $\widehat{F}_{\mathbf{W}}(\mathbf{w}) = \Phi(B)(1+o_P(1))$, and hence

$$\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} \xrightarrow{P} 1.$$

Thus,

$$\widehat{\mathrm{P}}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$= n_2 \widehat{F}_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1+N_2} + \widehat{R} - \left(n_2 F_{\mathbf{W}}(\mathbf{w}) - \frac{N_2}{N_1+N_2} + R \right)$$
$$= n_2 \left(\widehat{F}_{\mathbf{W}}(\mathbf{w}) - F_{\mathbf{W}}(\mathbf{w}) \right) + \left(\widehat{R} - R \right) \xrightarrow{P} 0,$$

noting that $n_2 \left(\widehat{F}_{\mathbf{W}}(\mathbf{w}) - F_{\mathbf{W}}(\mathbf{w}) \right) = n_2 F_{\mathbf{W}}(\mathbf{w}) \left(\frac{\widehat{F}_{\mathbf{W}}(\mathbf{w})}{F_{\mathbf{W}}(\mathbf{w})} - 1 \right) = o_P(1)$. □

1.10 Proof of Proposition 5.2

Proof. Let (U_1, V_1) and (U_2, V_2) be independent draws from C, and write

$$A_b = \{U_1 < U_2^b\}, \qquad B_a = \{V_1 > V_2^a\}, \qquad D = A_1 \cap B_1.$$

The paradox event is $E = A_b \cap B_a$. Because the two pairs are iid and their margins are continuous, the two orientations of a discordant pair have equal probability. Hence

$$\Pr(D) = \frac{1}{2} \Pr\{(U_1 - U_2)(V_1 - V_2) < 0\} = \frac{1-\tau}{4} = d_\tau.$$

Uniformity of the margins also gives

$$\Pr(A_b) = \int_0^1 u^b du = p_U, \qquad \Pr(B_a) = \int_0^1 (1 - v^a) dv = p_V.$$

If $b \le 1$, then $A_1 \subseteq A_b$ and $\Pr(A_b \setminus A_1) = e_U$; if $b \ge 1$, then $A_b \subseteq A_1$ and $\Pr(A_1 \setminus A_b) = c_U$. Similarly, if $a \ge 1$, then $B_1 \subseteq B_a$ with added probability e_V, whereas if $a \le 1$, then $B_a \subseteq B_1$ with removed probability c_V. Therefore,

$$D \setminus E \subseteq (A_1 \setminus A_b) \cup (B_1 \setminus B_a), \qquad E \setminus D \subseteq (A_b \setminus A_1) \cup (B_a \setminus B_1),$$

which implies

$$d_\tau - c_U - c_V \le \Pr(E) \le d_\tau + e_U + e_V.$$

The Fréchet bounds for $E = A_b \cap B_a$ give

$$\max(0, p_U + p_V - 1) \leq \Pr(E) \leq \min(p_U, p_V).$$

Intersecting the two intervals proves the stated bounds. If $a = b = 1$, all four adjustment terms vanish and $E = D$, so both bounds equal d_τ. If $\tau = 1$ and $b \leq 1 \leq a$, then $c_U = c_V = 0$ and $e_U + e_V = p_U + p_V - 1$, so the bounds again coincide. □

2 Simulations

2.1 Simulation Results for Theorem 4.4

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.4 in the numerical simulation. Paradox probabilities computed from Theorem 4.4 are compared with Monte Carlo estimates across a range of configurations. Each of these configurations will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.4 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation	Theory
10	40	400	600	-1	1	1.0	3.0	0.399932	0.400000
10	20	400	600	0	0	1.0	3.0	0.396838	0.397066
10	40	400	300	-1	1	1.0	3.0	0.571408	0.571429
10	20	400	600	0	1	1.5	2.0	0.364756	0.364746
10	40	400	600	0	1	1.0	3.0	0.400195	0.399998
20	40	400	600	0	1	1.0	3.0	0.400089	0.399996
10	40	400	600	-1	0	1.5	3.0	0.399687	0.399554
20	40	400	600	0	0	1.5	2.0	0.296258	0.296294
20	40	400	600	0	0	1.0	3.0	0.399977	0.399845
10	20	400	600	-1	1	1.0	3.0	0.400088	0.399991

Table 1: Simulation results for the normal maximum example. Each Monte Carlo estimate uses 10^7 independent draws. The theoretical probability is evaluated by one-dimensional numerical integration.

2.2 Sensitivity Analysis for Theorem 3.4

We extend the sensitivity analysis in the main text by varying the achiever rank $k \in \{0, 1, 2, 3\}$, where $k = 0$ corresponds to the largest achiever and $k = 3$ to the fourth-largest achiever. For each k, we consider the paradox event

$$Y_{1,(n_1-k)} < Y_{2,(n_2-k)} \quad \text{and} \quad X_{1,(N_1-k)} > X_{2,(N_2-k)},$$

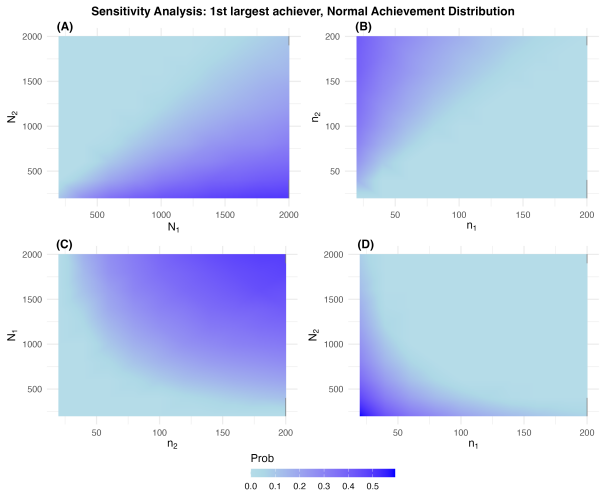
under the same marginal-distribution setting as Theorem 3.4: $X \sim \text{Pareto}(1.16)$ and $Y_1 \overset{d}{=} Y_2 \overset{d}{=} Y$. As in the main text, latent aptitudes are obtained through the order-statistic probability transformation (2). Each surface point is estimated using 10^4 Monte Carlo replicates.

In addition to the Normal achievement distribution $Y \sim \mathcal{N}(0, 1)$, we also consider a right-skewed achievement distribution $Y \sim \text{Gamma}(\alpha = 2, \beta = 1)$ to assess robustness to skewness. The four sensitivity panels (A)–(D) match those in the main text: (A) varies (N_1, N_2) with (n_1, n_2) fixed; (B) varies (n_1, n_2) with (N_1, N_2) fixed; (C) varies (n_2, N_1) with (n_1, N_2) fixed; (D) varies (n_1, N_2) with (n_2, N_1) fixed.

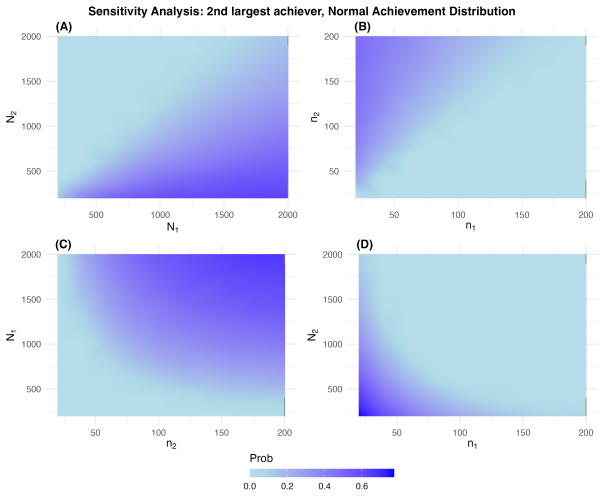
Figures 1–2 report the full set of surfaces for $k = 0, 1, 2, 3$ under $Y \sim \mathcal{N}(0, 1)$ and $Y \sim \text{Gamma}(\alpha = 2, \beta = 1)$, respectively. Across all k and both achievement distributions,

the qualitative dependence on (n_1, n_2, N_1, N_2) remains stable: increasing n_2 relative to n_1 increases the probability of the observed-achievement reversal, while increasing N_1 relative to N_2 increases the probability of the latent-aptitude reversal. As a result, the paradox probability is consistently largest when these two effects reinforce each other (large n_2/n_1 and large N_1/N_2), and smallest when they oppose each other.

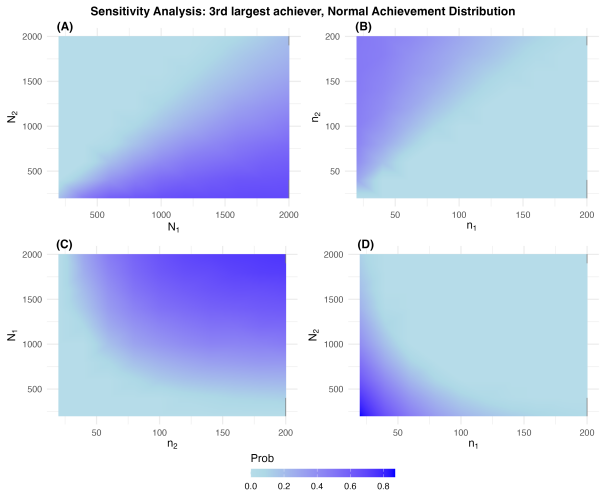
Varying k primarily changes the overall magnitude of the paradox probability, but it does not alter the location of high-probability regimes in the (n_1, n_2, N_1, N_2) design. Replacing $Y \sim \mathcal{N}(0, 1)$ by the skewed Gamma distribution, $Y \sim \text{Gamma}(\alpha = 2, \beta = 1)$, leads to similar surfaces in shape, indicating that the qualitative mechanisms are not driven by symmetry of the achievement distribution.



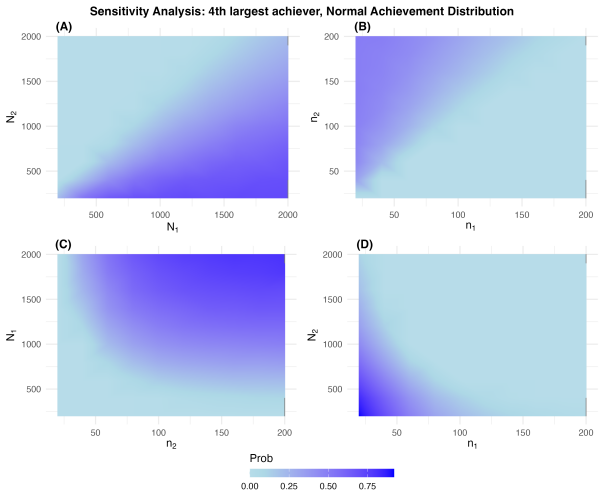
(a) $k = 0$ (largest achiever)



(b) $k = 1$ (second-largest)

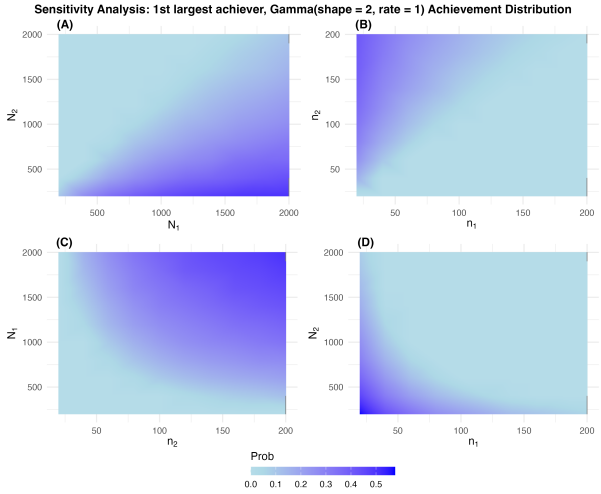


(c) $k = 2$ (third-largest)

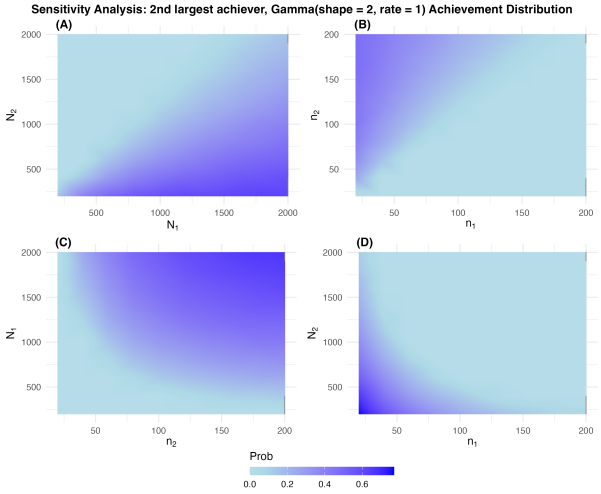


(d) $k = 3$ (fourth-largest)

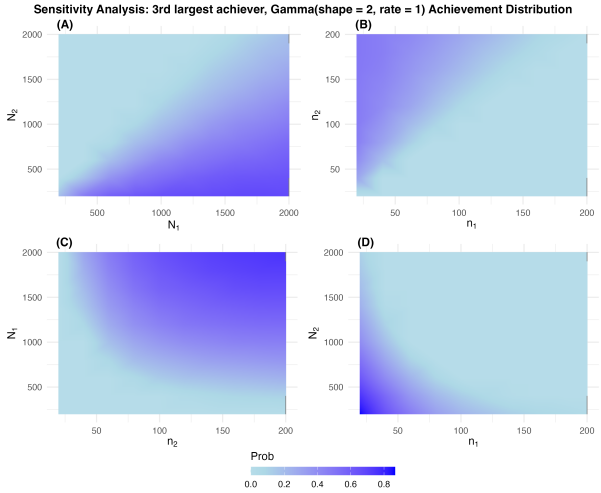
Figure 1: Sensitivity surfaces under $Y \sim \mathcal{N}(0, 1)$ for $k = 0, 1, 2, 3$ (largest through fourth-largest achiever). Each subfigure contains Panels (A)–(D) as in Figure 1. The darker color indicates larger paradox probability.



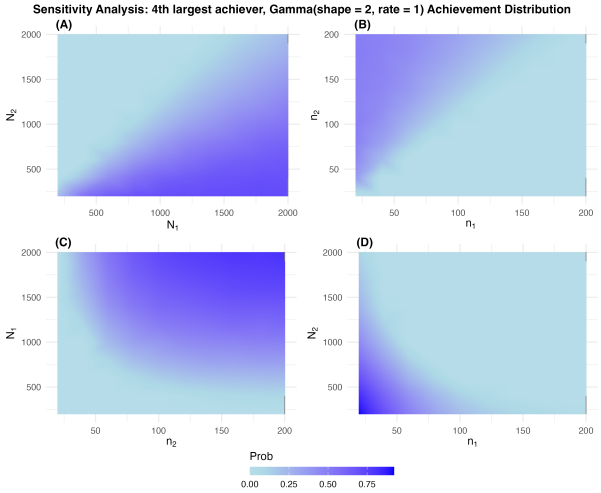
(a) $k = 0$ (largest achiever)



(b) $k = 1$ (second-largest)



(c) $k = 2$ (third-largest)



(d) $k = 3$ (fourth-largest)

Figure 2: Sensitivity surfaces under $Y \sim \text{Gamma}(2, 1)$ for $k = 0, 1, 2, 3$ (largest through fourth-largest achiever). Each subfigure contains Panels (A)–(D) as in Figure 1. The darker color indicates larger paradox probability.

2.3 Ratio of N_2 and N_1 Under a Skewed Gamma Achievement Distribution

We repeat the analysis of Section 6.2 (same (n_1, n_2), N_1 grid, and c grid) but replace the Normal achievement distribution by $Y \sim \text{Gamma}(\alpha = 2, \beta = 1)$, which is right-skewed. The paradox probabilities are estimated using 10,000 Monte Carlo repetitions for each configuration, and we report results for $k = 0, 1, 2, 3$ (largest through fourth-largest achiever).

Figure 3 shows that the dependence on c is qualitatively the same as in the Normal case: the paradox probability decreases as c increases, and conditional on c it exhibits little sensitivity to N_1 over the range considered. Thus, the ratio effect described in the main text is robust to moderate skewness in the achievement distribution.

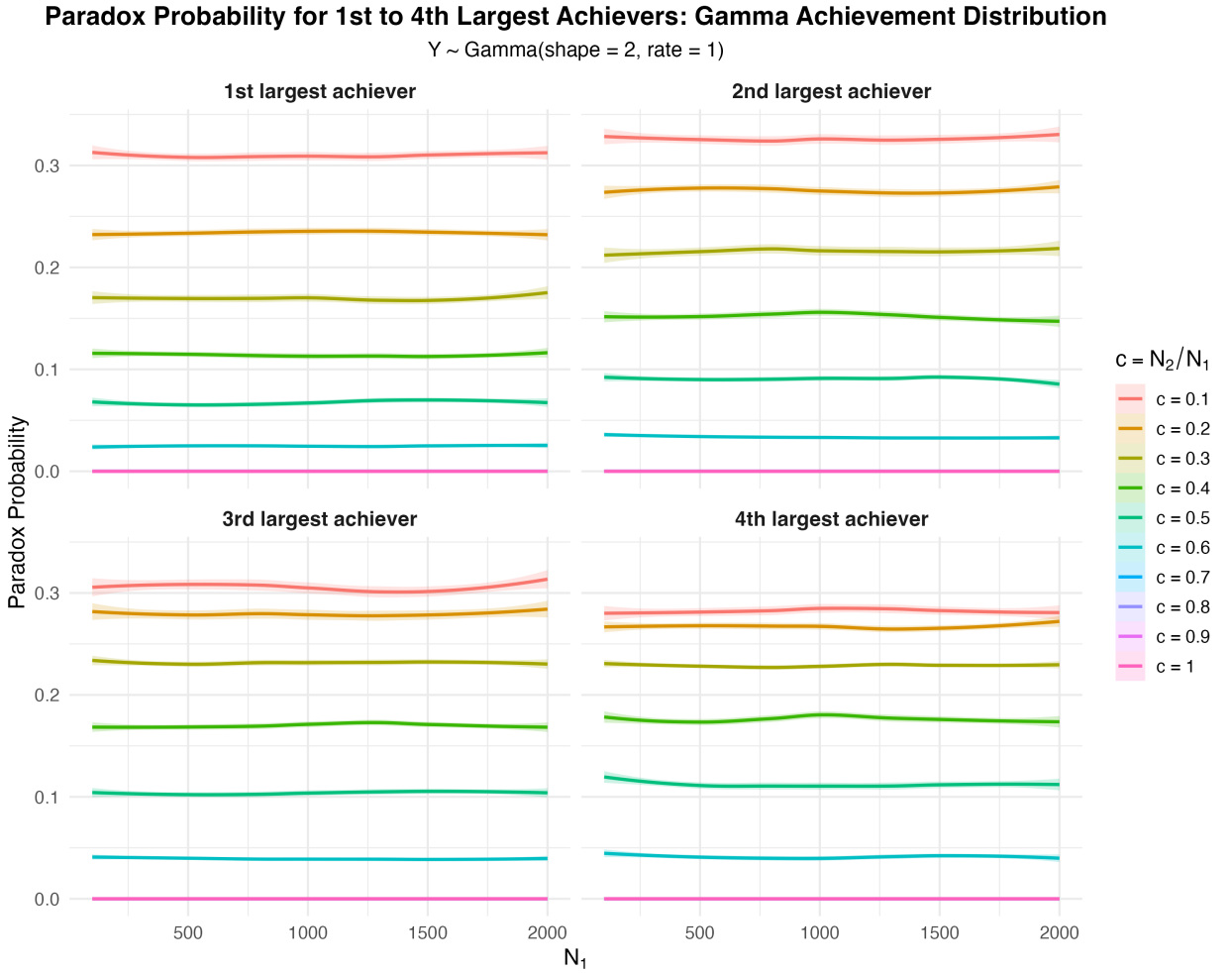


Figure 3: Gamma analogue of Figure 2 with $Y \sim \text{Gamma}(\alpha = 2, \beta = 1)$. The qualitative dependence on $c = N_2/N_1$ is unchanged.

References

Dimitris Askitis. Logarithmic concavity of the inverse incomplete beta function with respect to the first parameter. *Dimitris Askitis*, 2021.

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling extremal events: for insurance and finance*, volume 33. Springer Science & Business Media, 2013.

Alan Genz and Frank Bretz. *Computation of Multivariate Normal and t Probabilities*. Lecture Notes in Statistics. Springer-Verlag, Heidelberg, 2009. ISBN 978-3-642-01688-2.